

Elliptical Orbits and Kepler's Second Law

Elliptical Orbits & Kepler's 2nd Law · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/f6.html>

Curriculum links: gravity

Learning intentions

- I can describe how a planet's orbital shape depends on its initial speed and distance from the sun.
- I can explain Kepler's second law (equal areas in equal times) using the simulation's wedge overlay.

Getting started

Open the simulation and set up a circular orbit by choosing an initial speed and distance combination that produces a stable circle. Run the simulation to confirm.

Tasks

1. Starting from your circular orbit, increase the initial speed slightly and run the simulation. Describe the new orbital shape.
2. Keep increasing the initial speed in steps until the orbit becomes parabolic, then hyperbolic. At roughly what speed does the planet stop returning to the sun?
3. Return to an elliptical orbit and turn on the Kepler's 2nd law wedge display. Compare the size and shape of a wedge swept out near the sun (perihelion) to one swept out far from the sun (aphelion).
4. Using the wedges from Task 3, explain why the planet must be moving faster when it is close to the sun than when it is far away.

Extension (optional)

Earth's orbit is a (very slight) ellipse. Use what you found in Task 4 to explain why Earth moves fastest in early January, when it is closest to the Sun.

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